

Classification Model Comparison for Loan Approval Prediction: A Supervised Learning Approach

Abstract

Loan approval prediction is a critical task for financial institutions seeking to minimize default risk while maximizing profitable lending opportunities. This study compares five supervised machine learning models, Logistic Regression, K-Nearest Neighbors (KNN), Decision Tree, Random Forest, and Neural Network, for predicting loan approval status using a publicly available Kaggle dataset of 614 loan applicants with 13 features. Following standard data preprocessing (handling missing values, encoding categorical variables, and feature scaling), models were evaluated on accuracy, precision, recall, and F1-score using a 70/30 train-test split. Results demonstrate that Logistic Regression achieved the highest accuracy (84.86%) and recall (97.64%), outperforming ensemble methods, including Random Forest (83.78%). The Neural Network performed the worst (75.68% accuracy). Confusion matrix analysis revealed that Logistic Regression minimized false negatives (only 3 misclassified approved loans), a critical advantage for real-world lending where denying a qualified applicant carries lower costs than approving a defaulter. This research contributes empirical evidence that simpler linear models may outperform complex architectures for imbalanced tabular financial data, with direct implications for cost-sensitive business decision-making.

Keywords: loan prediction, classification, Logistic Regression, Random Forest, supervised learning, financial risk assessment

1. Introduction

1.1 Background of Customer Loan Prediction

The consumer lending industry represents a multi-trillion dollar global market where banks, credit unions, and fintech companies disburse millions of loans annually. The fundamental challenge facing lenders is asymmetric information: applicants know their own repayment likelihood, but lenders must infer this from limited application data. Traditional credit scoring methods, such as logistic regression on FICO scores and debt-to-income ratios, have served the industry for decades. However, the proliferation of digital lending platforms and alternative data sources has created both opportunities and demands for more sophisticated predictive techniques.

Loan prediction refers to the binary classification task of determining whether a loan applicant will repay their obligation (approved) or default (denied). This decision directly impacts institutional profitability: false approvals (Type I errors) generate non-performing assets and write-downs, while false denials (Type II errors) represent opportunity costs from foregone interest income and customer acquisition.

1.2 Real-World Impact

The consequences of loan prediction errors extend beyond balance sheets. For individuals, loan denial can restrict access to housing, education, and small business capital, perpetuating economic inequality. For institutions, the 2008 financial crisis demonstrated how systemic failures in risk assessment can trigger global recessions. Accurate loan prediction models therefore serve dual purposes: protecting lender solvency while expanding responsible credit access.

Modern lending decisions must also navigate regulatory frameworks including the Equal Credit Opportunity Act (ECOA) and Fair Credit Reporting Act (FCRA), which prohibit discrimination based on protected characteristics. This adds an interpretability requirement: models must not only predict accurately but also explain their decisions transparently.

1.3 Why Machine Learning Is Necessary

Traditional rule-based underwriting or simple threshold models (e.g., "approve if credit score > 650") fail to capture nonlinear interactions between features. For instance, high income may offset a short credit history, or large loan amounts may be safe for some professions but risky for others. Machine learning models automatically learn these complex patterns from historical data, potentially improving predictive accuracy over heuristic approaches. Furthermore, ML enables continuous learning as new loan performance data becomes available, allowing models to adapt to changing economic conditions.

1.4 Research Gap

While numerous studies have applied machine learning to credit scoring, several gaps persist. First, many published comparisons focus exclusively on accuracy without considering the asymmetric costs of misclassification in lending contexts. Second, ensemble methods like Random Forest and XGBoost are often presumed superior to simpler models without rigorous testing on imbalanced datasets. Third, relatively few studies compare neural networks directly with traditional classifiers on small-to-medium tabular datasets typical of individual lending institutions. This study addresses these gaps by evaluating five model classes with explicit attention to recall (minimizing false negatives) and by documenting performance on a realistic publicly available dataset.

1.5 Research Questions

Q1: Which machine learning model (Logistic Regression, KNN, Decision Tree, Random Forest, or Neural Network) performs best at predicting loan approval status based on accuracy, precision, recall, and F1-score?

Q2: What are the implications of the findings for business decision-making, particularly regarding the cost trade-offs between false approvals and false denials?

Q3: How can future studies improve loan prediction models through advanced techniques such as deep learning, oversampling, and interpretability methods?

2. Literature Review

This review synthesizes academic research on machine learning applications for loan prediction, customer analytics, and business decision modeling, identifying how the present study extends previous work.

Lessmann et al. (2015) conducted one of the most comprehensive comparative studies in credit scoring, evaluating 41 classification algorithms across eight real-world datasets. Their findings challenged the assumption that complex ensembles uniformly outperform simpler models while Random Forest and boosted trees performed well, logistic regression remained competitive, particularly when datasets contained missing values or nonlinear relationships were weak. The present study replicates their comparative framework but focuses on loan-specific features (credit history, property area) and provides explicit confusion matrix analysis for business decision-making.

Baesens et al. (2003) systematically compared neural networks, decision trees, and logistic regression for credit scoring, concluding that no single algorithm dominates across all datasets. They emphasized that model selection should consider implementation constraints (interpretability vs. predictive power) and data characteristics. This study extends their work by including KNN (absent from their comparison) and by using modern Python implementations rather than proprietary software.

Chen & guestrin (2016) introduced XGBoost, which has become a benchmark for tabular data competitions. While XGBoost was not included in this study (due to computational constraints in the base assignment), the present results establish a baseline against which future XGBoost comparisons could be made. The strong performance of Random Forest (83.8% accuracy) suggests that gradient boosting might further improve results.

Hand & Henley (1997) developed the concept of "reject inference"—the challenge that training data for credit scoring is inherently biased because only approved applicants have known outcomes. This study acknowledges this limitation (discussed in Section 7) but does not address it methodologically, representing a clear extension opportunity.

Khandani et al. (2010) applied classification trees to consumer credit risk using administrative data from a major bank, demonstrating that tree-based methods can reduce default prediction errors by 10-15% compared to logistic regression. Their work focused on interpretability through variable importance rankings. The present study extends this by comparing multiple tree variants (single Decision Tree vs. Random Forest) and documenting how ensembling improves generalization.

Siddiqi (2012) provided the industry-standard textbook treatment of credit risk scorecards, emphasizing that logistic regression remains dominant in regulated lending because it produces interpretable coefficients and satisfies fairness audits. The exceptional performance of Logistic Regression in our results (84.86% accuracy) corroborates Siddiqi's practical wisdom: for many lending applications, the interpretability-simplicity trade-off favors linear models.

Research Gap Addressed: While prior work has compared 2-3 model classes or focused on proprietary bank data, this study provides a reproducible comparison of five diverse algorithms on a public dataset with explicit discussion of business implications. Unlike many academic comparisons that optimize only accuracy, we prioritize recall (97.64% for Logistic Regression) to reflect real-world cost structures where false negatives (denying good loans) are less costly than false positives (approving bad loans).

3. Methodology

3.1 Dataset Description

Data were obtained from the Kaggle "Loan Prediction Problem Dataset" (altruistdelhite04, 2020), containing 614 loan applications with 13 features. The dataset represents anonymized applicants from a financial institution, with no geographic or temporal identifiers.

Target Variable: `Loan\_Status` (Y = approved, N = not approved). Distribution: 422 approved (68.7%), 192 denied (31.3%), indicating moderate class imbalance.

3.2 Variables (Features)

Variable Name	Type	Description
Loan_ID	Identifier	Unique loan identifier (dropped)
Gender	Categorical	Male/Female
Married	Categorical	Yes/No
Dependents	Categorical	0, 1, 2, 3+
Education	Categorical	Graduate/Not Graduate
Self_Employed	Categorical	Yes/No
ApplicantIncome	Numerical	Monthly income (USD)
CoapplicantIncome	Numerical	Co-applicant monthly income
LoanAmount	Numerical	Requested loan amount (thousands USD)
Loan_Amount_Term	Numerical	Repayment term (days)
Credit_History	Numerical	1 = good history, 0 = no/poor history
Property_Area	Categorical	Urban/Semiurban/Rural

3.3 Data Cleaning and Preprocessing

Missing Value Handling: Missing values were identified in Gender (13), Married (3), Dependents (15), Self_Employed (32), LoanAmount (22), Loan_Amount_Term (14), and Credit_History (50). Categorical missing values were imputed using the mode (most frequent value). Numerical missing values were imputed using the median to avoid distortion by outliers.

Encoding: The target variable was binary encoded (Y=1, N=0). All categorical features were converted using one-hot encoding with drop_first=True to avoid multicollinearity, resulting in 14 final features.

****Feature Scaling:**** StandardScaler (zero mean, unit variance) was applied to all numerical features for KNN and Neural Network models, which are distance-sensitive. Logistic Regression, Decision Tree, and Random Forest were trained on unscaled data as they are scale-invariant.

3.4 Train/Test Split

Data were split 70/30 training/testing using `train\_test\_split` with `random\_state=42` for reproducibility. Stratification was applied to preserve the original class balance (68.7% approved) in both splits. Final training set: 429 observations; testing set: 185 observations.

3.5 Explanation of Algorithms

Model	Description	Key Hyperparameters
Logistic Regression	Linear model estimating probability of approval via sigmoid function	max_iter=1000, C=1 (default)
K-Nearest Neighbors	Instance-based: class assigned by majority vote of k nearest neighbors	k=5 (tuned to 10)
Decision Tree	Recursive partitioning maximizing information gain at each split	max_depth=5, random_state=42
Random Forest	Ensemble of 200 decision trees with bootstrap aggregation	n_estimators=200 (tuned to 100), max_depth=5
Neural Network (MLP)	Multi-layer perceptron with 50 and 25 hidden neurons, ReLU activation	hidden_layer_sizes=(50,25), max_iter=1000

3.6 Evaluation Metrics

Accuracy: $(TP + TN) / (TP + TN + FP + FN)$ — overall correct predictions

Precision: $TP / (TP + FP)$ — of predicted approvals, how many were correct

Recall (Sensitivity): $TP / (TP + FN)$ — of actual approvals, how many were caught

F1-Score: Harmonic mean of precision and recall — balances both

Confusion Matrix: TP, TN, FP, FN counts enabling error analysis

ROC-AUC (recommended for future work): Area under the ROC curve measuring discriminative ability

3.7 Tools Used

All analyses were conducted in Python 3.9 using:

pandas(data manipulation)

numpy (numerical operations)

scikit-learn (preprocessing, models, metrics)

matplotlib (visualization)

Jupyter Notebook (reproducible environment)

Code and results are available in the companion notebook (classification_model_comparison_v2.ipynb).

4. Results

4.1 Accuracy Table (All Models)

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	0.8486	0.8322	0.9764	0.8986
Random Forest	0.8378	0.8255	0.9685	0.8913
KNN (k=10, tuned)	0.8162	0.8121	0.9528	0.8768
KNN (k=5, original)	0.7892	0.8099	0.9055	0.8550
Decision Tree (tuned)	0.8000	0.8169	0.9134	0.8625
Decision Tree (original)	0.7730	0.8148	0.8661	0.8397
Neural Network (tuned)	0.7892	0.8284	0.8740	0.8506
Neural Network (original)	0.7568	0.8254	0.8189	0.8221

Key Finding: Logistic Regression achieved the highest accuracy (84.86%) and recall (97.64%), correctly identifying 124 of 127 actually approved loans in the test set.

4.2 Confusion Matrices for Each Model (Tuned Versions)

Logistic Regression (Best Overall)

Confusion Matrix:

```
[[ 33 25] # TN=33, FP=25
 [  3 124] # FN=3, TP=124
```

False Negatives (FN=3): Only 3 approved loans incorrectly denied

False Positives (FP=25): 25 denied loans incorrectly approved (bigger business cost)

Random Forest

Confusion Matrix:

```
[[ 32 26]
 [  4 123]]
```

FN=4 (slightly worse than Logistic Regression)

FP=26

KNN (k=10)

Confusion Matrix:

```
[[ 30 28]
 [  6 121]]
```

FN=6 (twice Logistic Regression's FN rate)

Decision Tree

Confusion Matrix:

```
[[ 32 26]
 [ 11 116]]
```

FN=11 (substantially worse at identifying approved loans)

Neural Network

Confusion Matrix:

```
[[ 35 23]
 [ 16 111]]
```

FN=16 (worst performance on approved loans)

4.3 Feature Importance Plot (Random Forest)

[Plot generated from trained Random Forest classifier]

Top 5 Most Important Features (Gini Importance):

Feature	Importance Score
Credit_History	0.342
ApplicantIncome	0.187
LoanAmount	0.143
CoapplicantIncome	0.098
Property_Area_Semiurban	0.067

Interpretation: Credit history alone accounts for over one-third of predictive power. This aligns with domain knowledge: past repayment behavior is the strongest indicator of future default risk. Income and loan amount (capturing debt-to-income ratio implicitly) comprise the next most important factors. Property area suggests geographic variation in lending risk, possibly reflecting regional economic conditions.

5. Discussion

5.1 Which Model Performed Best and Why?

Logistic Regression achieved superior performance (84.86% accuracy, 97.64% recall) despite being the simplest model. Three factors explain this outcome:

1. Near-Linear Separability: The relationship between features (particularly credit history and income) and loan approval appears approximately linear. When decision boundaries are linear,

logistic regression's parametric form is optimally efficient, whereas nonlinear models (neural networks, decision trees) may overfit noise.

2. Imbalanced Class Handling: Logistic regression's probabilistic output allows easy threshold adjustment (though we used the default 0.5). Its high recall suggests the learned coefficients strongly weight `credit_history=1` toward approval predictions.

3. Small Sample Size: With only 614 observations, complex models with many parameters (neural network: ~1,500 weights) suffer from high variance. Logistic regression's parsimony (14 coefficients) provides regularization by design.

The Neural Network performed worst (75.68% accuracy), likely due to insufficient training data; deep learning typically requires thousands of examples to outperform simpler methods on tabular data.

5.2 Trade-offs: Accuracy vs. Interpretability

Model	Accuracy	Interpretability	Business Suitability
Logistic Regression	84.9%	High (coefficient weights)	Excellent for regulated lending
Random Forest	83.8%	Medium (feature importance)	Good for internal risk models
Decision Tree	77.3%	High (rule extraction)	Useful for policy explanation
Neural Network	75.7%	Low (black box)	Poor for compliance needs

For lending decisions subject to ECOA/Fair Lending regulations, interpretability is not optional; institutions must explain adverse actions to applicants. Logistic regression's coefficient weights (e.g., "having credit history increases approval odds by 4.2x") satisfy this requirement. Random Forest offers partial interpretability via feature importance but cannot produce individual-level explanations without additional tools (SHAP/LIME).

5.3 What Variables Were Most Important?

Credit history dominated all models' predictions. Applicants with a credit history (`Credit History=1`) had dramatically higher approval probabilities; this variable alone may approach a "silver bullet" feature in this dataset. Income and loan amount followed, capturing the ability to repay. Dependents and education showed negligible importance, suggesting the dataset may lack variation on these dimensions or that the institution's underwriting criteria ignore them.

Practical Implication: Institutions lacking credit history data (e.g., thin-file applicants) would see severe model degradation. Alternative data (rental payments, utility bills) would be necessary for financial inclusion.

5.4 Business Implications (Cost of Misclassification)

In lending, Type I errors (false positives: approving a defaulter) carry direct costs equal to the loan amount written off. Type II errors (false negatives: denying a good loan) carry opportunity costs equal to lost interest income (typically 5-15% of principal).

For a hypothetical \$10,000 loan:

False Positive Cost: \$10,000 (loss of principal)

False Negative Cost: \$1,000 (foregone interest at 10% APR)

Thus, false positives are approximately 10x more costly than false negatives. Models should therefore be optimized for precision (minimizing FP) or a custom cost-sensitive metric. Our Logistic Regression model achieved a precision of 83.2% (of 149 approvals predicted, 124 were correct). The alternative KNN model (precision 81.2%) would generate 4-5 additional defaults per 100 approvals, potentially costing the institution \$40,000-\$50,000 per 100 loans.

Recommendation: Deploy Logistic Regression, but adjust the classification threshold from the default 0.5 to 0.7 or higher, trading off some recall for improved precision. For every 100 applications, this might reduce false approvals from 25 to 15, saving \$100,000 while denying 5 additional good applicants (costing \$5,000 in lost interest); this is a net benefit of \$95,000.

5.5 Ethical Considerations

Three ethical concerns require attention:

1. Fairness and Protected Classes: Features like Gender and Married status, while predictive, cannot legally be used for lending decisions in many jurisdictions. If the model learns that "married applicants have lower default rates," using this for approval would discriminate against unmarried individuals. A fairness-aware approach would either remove these variables or enforce demographic parity constraints.

2. Transparency: The "black box" problem is acute for neural networks. Applicants denied a loan have legal rights to an explanation. Logistic regression and decision trees can provide this (e.g., "denied because credit history=0"), but neural networks cannot without post-hoc approximations like SHAP, which remain contested.

3. Reproducibility of Harm: If historically biased lending data (e.g., redlining) is used for training, models will perpetuate and amplify discrimination. Our dataset's credit_history variable may itself reflect systemic biases—if minority communities had differential access to credit-building opportunities, the model will learn historical inequity as predictive signal.

Mitigation Strategy: Prior to deployment, conduct disparate impact analysis (80% rule) and equal opportunity difference testing across protected groups. Consider excluding demographic variables entirely, even if they improve accuracy.

6. Future Studies

While this study provides a robust comparison, several limitations suggest future research directions.

6.1 Advanced Modeling Techniques

Deep Learning (LSTM, ANN): Our simple MLP underperformed due to limited data. Future work with larger datasets (10,000+ loans) could evaluate deeper architectures, including LSTM networks if temporal application sequences exist (e.g., repeated applications from the same customer).

Gradient Boosting (XGBoost, LightGBM, CatBoost): These ensemble methods often outperform Random Forest on tabular data. XGBoost's ability to handle missing values natively and learn from residuals could potentially push accuracy above 88%.

6.2 Addressing Class Imbalance

Our dataset was imbalanced (68.7% approved). Future work should implement:

SMOTE (Synthetic Minority Oversampling): ** Generate synthetic examples of denied loans to balance training data. This often improves recall for the minority class.

Cost-Sensitive Learning: ** Assign misclassification costs directly to the loss function (e.g., FP cost = 10, FN cost = 1). This would explicitly optimize for business value rather than accuracy.

Threshold Tuning: Systematically find optimal decision threshold using profit curves rather than the default 0.5.

6.3 Feature Engineering

Current features are raw application fields. Engineered features could improve performance:

Debt-to-Income Ratio: $\text{LoanAmount} / (\text{ApplicantIncome} + \text{CoapplicantIncome})$

Loan Amount per Dependent: $\text{LoanAmount} / \text{Dependents}$ (capturing household burden)

Income Stability Proxy: $\text{Self_Employed} \times \text{ApplicantIncome}$ (interaction term)

Geographic Risk Scores: Replace `Property_Area` with regional default rates from external data

6.4 Temporal and Behavioral Data

The current dataset is cross-sectional. Adding time-series data (e.g., past 12 months of bank transactions, payment histories on existing loans) would enable:

Survival Analysis: Predict not just default but time to default

Recurrent Neural Networks: Model sequences of customer behavior

Early Warning Systems: Detect deteriorating financial health

6.5 Customer Satisfaction Integration

Loan prediction typically ignores customer experience. Future research could incorporate post-approval survey data (satisfaction scores, Net Promoter Score) as either:

Additional feature (e.g., "highly satisfied customers have lower default rates")

Multi-objective optimization** (maximize approval accuracy and customer satisfaction)

6.6 Interpretability Methods

To address regulatory concerns, future studies should systematically compare:

SHAP (SHapley Additive Explanations): Game-theoretic feature attribution

LIME (Local Interpretable Model-agnostic Explanations): Local surrogate models

Anchors: High-precision rule-based explanations

Counterfactuals: "To be approved, your income would need to be \$X"

These methods could be applied to our top-performing Logistic Regression and Random Forest models to produce compliant adverse action notifications.

6.7 External Validation

Our results are dataset-specific. Future work should replicate this comparison on:

Multiple lending datasets (microfinance, mortgage, auto loans)

Different time periods (pre/post COVID-19 recession)

Different geographic regions (developing vs. developed economies)

This would establish whether Logistic Regression's dominance generalizes or is unique to this dataset.

7. Conclusion

This study compared five supervised classification models—Logistic Regression, K-Nearest Neighbors, Decision Tree, Random Forest, and Neural Network—for the task of predicting loan approval status using a real-world dataset of 614 applicants. Following systematic preprocessing, missing value imputation, and encoding, models were evaluated on accuracy, precision, recall, and F1-score with a 70/30 stratified train-test split.

Logistic Regression achieved the highest accuracy (84.86%) and recall (97.64%), correctly identifying 124 of 127 approved loans while misclassifying only 3. Random Forest performed comparably (83.78% accuracy), while simpler KNN (78.92%) and Neural Network (75.68%) lagged substantially. Confusion matrix analysis revealed that Logistic Regression minimizes false negatives, which is crucial in lending, where denying a qualified applicant (opportunity cost) is preferable to approving a defaulter (direct loss). Feature importance from Random Forest identified credit history as the dominant predictor (34.2% of total importance), followed by applicant income and loan amount.

The primary contribution of this study is empirical evidence that for small-to-medium imbalanced tabular financial datasets, linear models may outperform complex ensembles and neural networks. This challenges the assumption that "more complex equals better" and has direct business implications: financial institutions facing regulatory interpretability requirements can deploy Logistic Regression with confidence, achieving both high performance and compliance. The cost analysis demonstrates that optimizing for recall (minimizing false negatives) aligns with business objectives where false positives (approving defaulters) carry 10x higher costs.

Key lessons learned from comparing multiple algorithms include: (1) no single model dominates universally—performance depends on dataset size, class balance, and linearity; (2) ensemble methods reduce overfitting but may not improve accuracy when the true decision boundary is simple; (3) confusion matrix error analysis provides more actionable insights than accuracy alone for cost-sensitive applications; and (4) interpretability requirements in regulated industries may eliminate black-box models from consideration regardless of predictive performance.

Limitations include the modest dataset size (614 observations), lack of temporal validation, and absence of fairness auditing. Future work should address these through gradient boosting methods (XGBoost), SMOTE oversampling, fairness constraints, and SHAP interpretability tools. Despite these limitations, this study provides a reproducible methodology and baseline results that advance the practical application of machine learning to consumer lending.

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